

11.8 LET US SUM UP

In this unit, we discussed about foreign exchange market and international financial market instruments. Foreign exchange market refers to instruments denominated in foreign currencies with a maturity of different periods from one day to more than a year. The foreign exchange market is dominated by many central banks, commercial banks, other financial institutions, and nonfinancial institutions, such as corporations and governments. All these are players in the foreign-exchange market. The foreign-exchange market is divided into the over-the-counter market (OTC) and the exchange-traded market. Foreign exchange is money denominated in another nation's currency or group of nations. The exchange rate is the price of a currency. Companies use foreign exchange to settle transactions involving the imports and exports of goods and services, for foreign investments, and to earn money through arbitrage or speculation.

International financial markets and operations comprise buying and selling currencies, deposit taking and lending and issuance of securities. Market segments are classified according to the nature of financial operations. Money markets deal with exchange or exchange-related transaction, credit markets deals with deposit taking and lending. Capital markets and equity markets are concerned with issuance of securities and issuance of international equities respectively.

An option is the right, but not the obligation, to trade foreign currency in the future. Options can be traded OTC or on an exchange. A foreign currency future is an exchange-traded instrument that guarantees a future price for the trading of foreign exchange, but the contracts are for a specific amount and specific maturity date.

Euro market transactions are financial transactions denominated with currency outside the country of its origin. Euro markets consist of Eurobanks that offer wholesale deposits and loans in favourable jurisdiction and in a variety of currencies other than that of the country where the banks are situated. These Euro banks are entirely free of market regulatory controls. Euro market operations have been medium to long-term lending with floating interest rates. Indian companies have been raising funds from international financial markets by issuing Euro bonds, Euro-convertible bonds and Euro equities. 'Global Depository Receipts (GDRs) are used to tap multiple markets using a single instrument.

Risk is widespread in financial markets. The risk in international financial markets is greater than domestic markets. Various kinds of risk are involved in international financial transactions. Such risks are generally categorised into three: transaction risk, translation risk, and economic risk.

11.9 KEY WORDS

Euro banks	The banks which accept deposits and make loan in foreign currencies.
Euro bonds	A bond placed in the countries other than those in whose currency it is denominated.
Euro-Commercial Papers	These are short-term promissory notes issued on an unsecured basis by commercial and financial institutes.
Euro dollar	Dollar deposited in a bank outside its country of origin.
Euro market	Transactions in Eurodollar, Euroyen etc.
Foreign Bond	The bond sold outside the borrower's country and is denominated in the currency of the country of issue.
LIMEAN	The average of LIBID and LIBOR.
London-Interbank Offered rate (LIBOR)	The rate at which funds are offered to a London or by a London bank to another London Bank.

11.10 SOME USEFUL REFERENCES

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11.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See section 11.2
- 2) See section 11.3
- 3) See section 11.3.1

Check Your Progress 2

- 1) See section 11.4
- 2) See section 11.4
- 3) See section 11.5

Check Your Progress 3

- 1) See section 11.6
- 2) See section 11.6
- 3) See section 11.6
- 4) See section 11.7
- 5) See section 11.7

UNIT 12 FINANCIAL AND CURRENCY CRISES *

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Explaining Financial Crisis
- 12.3 Global Financial Crisis 2007
- 12.4 Unfolding of Global Financial Crisis
 - 12.4.1 Structure and the Role of Credit Derivatives
 - 12.4.2 Residential Mortgage-Backed Securities and the bubble burst
 - 12.4.3 How the crisis of 2007 began?
- 12.5 World's most Devastating Financial Crises in History
- 12.6 The Currency Crisis and Its Effects on Financial Markets
- 12.7 Causes of the Financial Crisis of 2008
 - 12.7.1 US Banking system problem
 - 12.7.2 Global Imbalances
 - 12.7.3 Growth of Securitization and the OTD model
 - 12.7.4 Financial Innovation
 - 12.7.5 US Monetary Policy
 - 12.7.6 Political Factors
- 12.8 The Effects of the Crisis on the Macroeconomy
 - 12.8.1 Credit Demand Effects
 - 12.8.2 Credit Supply Effects
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- 12.10 Let Us Sum Up
- 12.11 Key Words
- 12.12 Some Useful Readings
- 12.13 Answer/Hints to Check Your Progress Exercises

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12.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain the reason behind Financial and currency crisis;
- describe the causes, features, consequences, types, and implications of the Global Financial Crisis;
- analyse the structure and organisation of the Credit Derivative products and their role in the global financial crisis; and
- explain the regulatory responses to the Global Financial Crisis.

12.1 INTRODUCTION

In this unit, we will understand the phenomenon of financial crisis and its origin. You will also get to learn about the way through which the financial crisis affects any economy. Afterwards we will discuss the Global financial crisis of the year 2007 at length by describing its causes, features, consequences, types and implications. We also discuss the structure and organisation of Credit Derivative products and their role in the global financial crisis. Any financial crisis needs to be managed or combatted. To do so there are regulatory responses, which we will discuss in detail.

12.2 EXPLAINING FINANCIAL CRISIS

Financial System consists of financial markets, financial instruments, and financial services. The financial system plays a pivotal role in channelising savings from savers to dissavers. The Merriam-Webster dictionary defines a crisis as an unstable/crucial time or situation in which a decisive change is impending. A financial crisis is when financial asset prices are nosedived in value, firms, businesses, multilateral organisations and consumers cannot pay their debts, and financial institutions experience a shortage in liquidity. A financial crisis is often associated with a bank run where savers lose faith in the banks because of inadequate deposit insurance. A financial crisis is often associated with one or more phenomena: the bursting of a financial bubble, the mortgage crisis, sovereign default, currency crisis, stock market crashes, credit crunches, severe disruptions in financial intermediation and the supply of external financing to various sectors in the economy, large scale government support (in the form of liquidity support and recapitalisation. A financial crisis is sometimes limited to a single country or one segment of the financial services but is more like to spread regionally or globally. Thus, financial crises are typically multidimensional and can be hard to characterise by employing a single indicator.

There are two types of financial crises such as systemic financial crises and non-systematic financial crises. A systemic crisis refers to the risk of the breakdown

of an entire financial system rather than the failure of one isolated part. It denotes the risk of cascading failure in the financial sector caused by the vertical and horizontal integration within the financial system, as during the Great Depression in the 1930s and the global financial crisis in 2008. As a result, it leads to a severe economic downturn. On the contrary, a non-systemic crisis involves only one or a few markets or sectors, like the Savings and Loan Crisis. Apart from these, the other forms of financial crises are currency crises, capital account or balance of payment crises, debt crises and bank crises. A financial crisis occurs when policymakers and finance ministers ignore indirect warnings given by the country's financial market and cannot understand, estimate and forecast the associated risk. Deficiency in financial regulations and under supervision brings instability to the nation's financial markets.

Financial crises are age-old problems. The series starts with debt and inflation crises in ancient Greece and Rome to banking, debt, and currency crises affecting many countries in modern times. Each past crisis is an essential subject for policymakers, social scientists and financial practitioners as it teaches a lesson about how to handle and prevent future crises. Learning from past mistakes and corrective measures not only reduces the incidence and impact of the financial crisis but also enhances understanding of monetary circulations, the impact of internal and external shocks, exchange rate regimes, interest rate and macro linkages etc. – Irving Fisher, Milton Friedman, John Maynard Keynes, Paul Krugman, and John B. Taylor are famous examples.

The literature associated with the financial crisis opines that fundamental factors, such as macroeconomic imbalances in terms of shocks in money supply, output, prices, wages, interest rates, exchange rates, current account deficits, rising fiscal deficits, the balance of payments deficits in terms of volume and receipts of exports and imports, financial openness measures, trade openness indicators, foreign investment openness indicators are the potential factors behind the financial crisis. Financial crises sometimes arrive by irrational factors such as bank runs, contagion and spillovers among financial markets, limits to arbitrage during times of stress, the emergence of asset busts, credit crunches etc.

Even after the structural economic reforms in many countries, such as the advent of globalisation, privatisation, and liberalisation, now the question arises what effects do these structural reforms in the financial sector have on economic growth and vice versa? If the financial and real sectors are interlinked, why could macroeconomists and statisticians not forecast the financial crisis well in advance? Joseph Schumpeter (1911) was perhaps the first economist to examine this nexus. Schumpeter distinguishes between producers and entrepreneurs, where entrepreneurs take risks and go for new combinations in the form of innovation. In Schumpeter's theory of economic development, this unique combination or innovation is the main driver to take the economy from a lower level to a higher equilibrium through bank credit. However, in the modern

banking system, bank credit severely suffers because of a lack of regulatory control and a deposit insurance scheme to protect the banking system. The gross mismatch between short-term saving liabilities of banks for financing long-term investments and uncertainties in the financial markets leads to panic in the bank customers and bank runs.

Different schools of thought have multiple perspectives on the role of finance in economic growth. But it is widely accepted in the literature that the financial system plays a pivotal role in facilitating savings in terms of investment and contributing towards growth. However, this is subject to regulatory control and market discipline.

12.3 GLOBAL FINANCIAL CRISIS 2007

The global financial crisis, which began to manifest itself in July of 2007, emerged as a liquidity crisis, which can be traced back to the US subprime housing market. When interest rates for loans rose between 2004 and 2006, and the prices of houses began to decline in 2007, failures in the repayment of subprime mortgage loans, as well as failures in the repayment of subprime mortgage-backed debts and other securities, heavily increased. Major financial institutions that had borrowed or invested in mortgage-backed securities of non-depository finance companies were forced to declare bankruptcy or were taken over by other banks. The emergence of the global financial crisis was also the result of the doubtful use of high-risk off-balance sheet instruments with zero liquidity and uncertain value on the side of non-depository financial institutions. Mortgage-backed securities and collateralised debt obligations (CDO), traded by commercial banks to distribute credit risk, which turned out to be illiquid for most of the part, heavily accelerated in the year before the financial crisis surfaced.

By contagion, the housing market downturn and subsequent financial market crisis have also affected the broader US real economy. Due to the growing integration of financial markets and industries during the past 20 years, the crisis spread to Europe with dramatic slowdowns occurring across Europe and the recession widening into a global problem. Some of the factors that led to the financial crisis in the US – high-risk, non-banking mortgage loans, low-interest rates, aggressive (subprime) credit expansion, and ignoring of the credit risks of borrowers – created similar problems in Europe.

The US crisis affected almost every part of the world in the new, globalised world of closely interdependent economies. The global financial crisis highlighted the complex international linkages between financial markets and between the US and European economies.

12.4 UNFOLDING OF THE GLOBAL FINANCIAL CRISIS

The financial crisis in 2007 was a non-systemic crisis linked to subprime mortgages or risky loans to homeowners. The US subprime mortgage market delivered a significant shock to the existing architecture for financial services regulation and supervision worldwide. The foundation of the global financial crisis was the housing market bubble which was the result of a political mandate in the US wherein each homeless US citizen was to be provided with a house. The US Federal Reserve bank of St. Louis instructed all the commercial banks in the US territory to provide an easy loan to all US borrowers interested in financing their housing needs. However, commercial banks found difficulty due to the stringent restrictions like asset and income verification of the borrowers before sanctioning home loans to them. Historically, banks approve a loan if the value of the asset possessions is sufficiently higher than the value of the liabilities the borrowers asked for. Hence, potential homebuyers found it difficult to obtain mortgages if they had below-average credit histories, provided small down payments or sought high-payment loans. Lenders denied such mortgage requests unless protected by government insurance or guarantee schemes.

Nevertheless, with the instructions of the government mandate and the circular from the US Federal Reserve Bank, the commercial banks were asked to revise their lending rate at a lower rate to make the loan easier for home loan borrowers despite their low credit history. This is popularly known as 'Sub Prime Lending Based' home loans. Thus, high-risk mortgages became available from lenders who funded mortgages by repackaging them into pools sold to investors in mid 2003s.

A set of new financial products was used to appropriate these risks, with private-level mortgage-backed securities (PMBS) providing most of the funding for subprime mortgages. These instruments are called credit derivative instruments, financial reengineering products, securitisations, and asset-backed securities. Credit derivatives may also be defined as a specific class of financial instruments, the value derived from an underlying market value of the reference asset bank loan or a portfolio of loans, corporate bonds or other debt instruments, trade receivables etc. These instruments are Collateral Bond Obligations, Collateral Debt Obligations, Collateral Mortgage Obligations, Credit Default Swaps, Total Return Swaps, Credit Linked Notes, Residential Mortgage Based Securities, and Commercial Mortgage Based Securities. This enabled more first-time homebuyers to obtain mortgages and homeownership rose. In the next section, we have described a particular structure of Residential Mortgage Based Securities to understand the global financial crisis.

12.4.1 Structure and the Role of Credit Derivatives

To understand the structure and the role of credit derivatives in the subprime mortgage crisis, let's divide the banks into commercial or conventional and modern or investment banks. Commercial banks accept deposits from customers with non-maturing liabilities to finance the borrowers and investments with long-term maturities. However, investment banks manage the depositors' wealth on their behalf. Credit Derivatives (CDs) are off-balance sheet financial contracts that allow the lender to transfer the credit risk of a financial instrument to a third party against the payment of a fee. They are voluminously used in the US commercial banking sector. The banks use them to mitigate credit risk exposure and expand their credit portfolio. A credit derivative involves mainly three participants. First, the borrower needs credit (reference entity). Second, the lender needs insurance for the loan (protection buyer). Third, the third party or the guarantor who wants to profit by covering the risk of a loan default (protection seller / Counterparty). There are two types of credit derivatives. They are Funded and Unfunded credit derivatives. Unfunded CDs are contingent instruments where the protection buyer does not receive any initial payment from the protection seller (counterparty) for covering the credit risk. Under such an agreement, the counterparty pays only when the borrower defaults. The different types of unfunded CDs are Credit Default Swaps (CDS), Credit Default Swap Options (CDSO), Credit Spread Options (CSO), and Total Rate of return Swaps (TRRS).

On the other hand, funded CDs are instruments where the lender is not exposed to credit risk from this counterparty. This is because the counterparty pays the lender an appropriate sum to cover any future loan default. The different types of funded CDs are Credit Linked Notes (CLNs), Collateral Debt Obligations (CDOs), Collateral Mortgage Obligations (CMOS), Collateral Bond Obligations (CBOs), Commercial Mortgage-Backed Securities (CMBS), Residential Mortgage-Backed Securities (RMBS) and Constant Proportion Debt Obligation (CPDO).

12.4.2 Residential Mortgage-Backed Securities and the bubble burst

Providing easy loans to the homeless forced all the commercial banks in the US territory to disburse the home loan to customers despite their weak creditworthiness. The primary incentive before these commercial banks was that they generate these home loan portfolios for the next 15 years to 20 years of maturity so that they can sell the home loan portfolio to investment banks to enjoy the time value of the home loan portfolio without exposing to the credit risk even though home loan portfolio is risky. This is because any of the borrowers can default at any point in time for the next 15 to 20 years on their commitment while paying towards the principal and interest rate of their total loan value. The investment bank purchases this home loan portfolio and creates a

trustee to manage this portfolio. On the full value of this credit portfolio, the trustee also buys the credit protection in the form of insurance with the condition that the protection seller will compensate the amount if any borrower defaults in the credit home loan portfolio during the maturity of the loan. In anticipation, the trustee pays the regular premium to the credit protection seller during the contract's life. These protection sellers are the same investment banks that have created the trusts to manage the credit home loan portfolio. The trustee, also known as a special purpose entity, has taken some amount as reserve and cash account to utilise for the contingency purpose. Based on the remaining amount, otherwise known as asset-backed securities, the trustee has issued different financial instruments such as senior notes, mezzanine notes, junior notes and equity tranches to issue to the investors. There is a hierarchy among these notes, so senior notes provide less return with low risk compared to the mezzanine, junior and equity tranches. The equity tranche is highly risky and provides a higher return. However, it is difficult to find investors for equity tranches in the market; hence, the same investor bank is the investor for equity tranches. These notes are high investment graded by rating agencies such as Standard and Poor, Moodys, and Fitch. A typical structure of RMBS is given as follows.

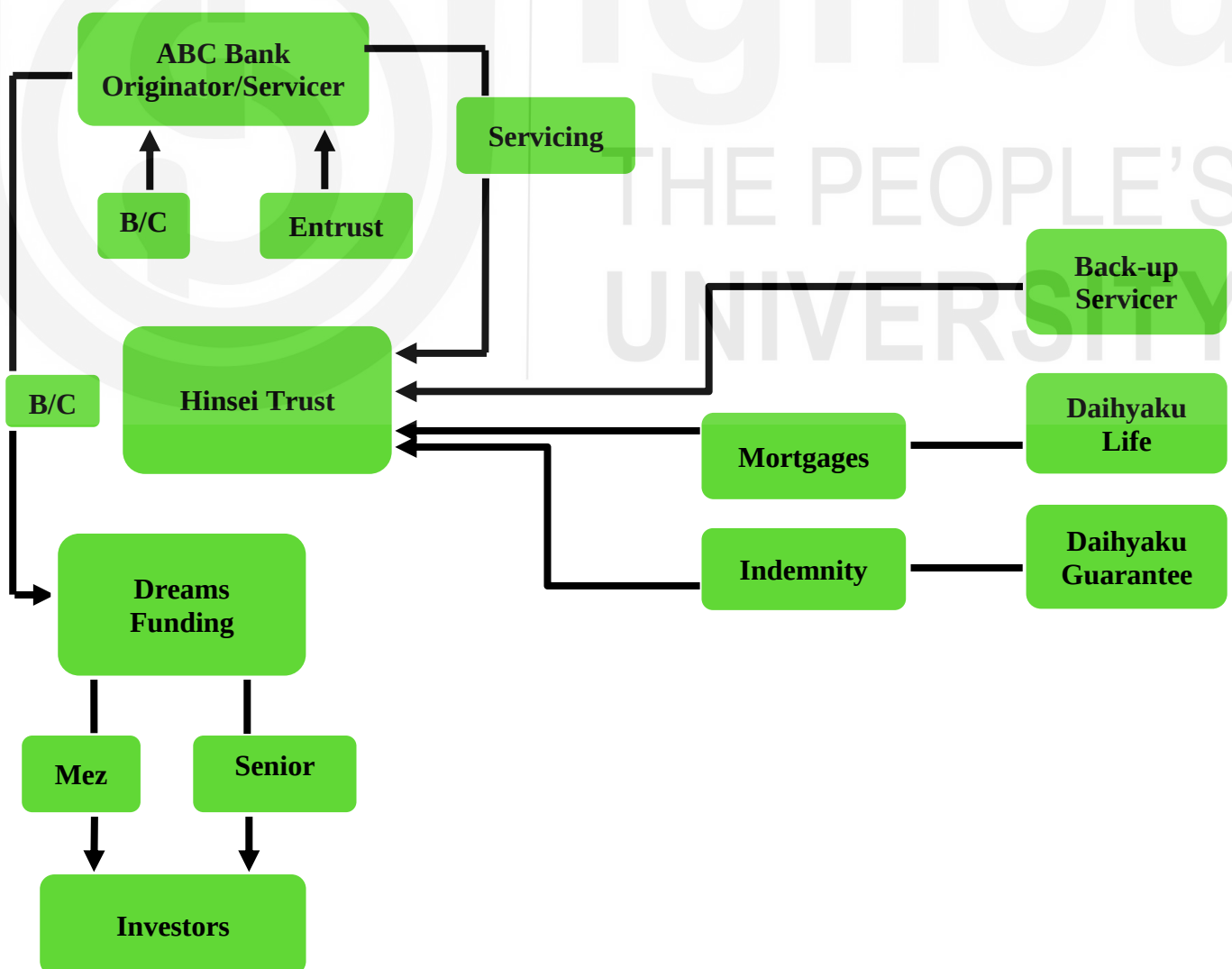


Fig. 12.1: Schematic Diagram of Residential Mortgage-Backed Securities